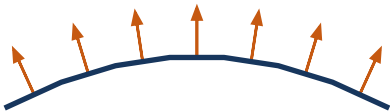
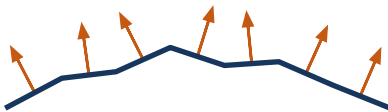


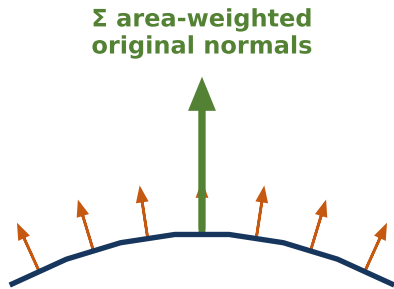
(a) Original support



(b) Repeated local updates



(c) Cluster-normal memory



Local face-to-face penalties can drift after many collapses; carrying the sum of original support normals preserves a stable reference.